

WEEK 6 PROJECT SUMMARY

1. What Was Planned

The objective was to improve the Week 5 no-show prediction baseline through error analysis, feature refinement, advanced modelling, cross-validation, threshold analysis and validation.

2. What Was Completed

Week 6 completed:

- ✓ Week 5 baseline reconstruction
- ✓ baseline performance review
- ✓ false-positive analysis
- ✓ false-negative analysis
- ✓ feature refinement
- ✓ improved Logistic Regression
- ✓ Random Forest modelling
- ✓ model comparison
- ✓ five-fold stratified cross-validation
- ✓ threshold analysis
- ✓ candidate model evaluation
- ✓ model interpretation
- ✓ business relevance assessment
- ✓ cross-track integration documentation
- ✓ risk and limitation assessment
- ✓ Week 7 testing requirements

3. What Was Improved From Week 5

The modelling process was extended beyond the initial baseline. Week 6 introduced broader feature evaluation, additional modelling approaches, cross-validation and threshold analysis.

4. What Was Integrated

The Data Science work was connected with the Data Analytics track so that analytical attendance findings could inform or validate predictive feature selection and interpretation.

5. Track Collaborated With

Data Analytics.

6. What Was Exchanged

Data Analytics provided attendance-related analytical findings, while Data Science provided model results, error analysis, refined features and candidate model requirements.

7. What Changed as a Result

The collaboration should result in a documented change to feature selection, model interpretation, validation or another modelling decision.

8. Key Findings / Development Outcomes

The Week 6 analysis provides evidence about:

- ✓ baseline model weaknesses
- ✓ the distribution of false positives and false negatives
- ✓ the effect of refined features
- ✓ differences between Logistic Regression and Random Forest
- ✓ cross-validation performance
- ✓ classification-threshold trade-offs

9. Major Challenges

Major challenges include limited data, potential generalisation issues, feature availability, class imbalance, threshold selection and the need to avoid target leakage.

10. Important Decisions

No-Show recall was treated as an important operational metric. Multiple models were compared rather than relying on accuracy alone. Threshold analysis was introduced to investigate the operational trade-off between identifying more potential No-Shows and generating additional false positives.

11. Remaining Issues

Further testing is required before the model can be considered suitable for operational use.

12. Contribution to HealthConnect

The Week 6 work advances HealthConnect from an initial predictive baseline toward a more thoroughly validated candidate model that can enter Week 7 testing.

13. Week 7 proposed work

The candidate model should be tested further in Week 7.

Testing should include:

- ✓ Evaluation on unseen data.
- ✓ False-positive and false-negative testing.
- ✓ Prediction threshold analysis. Model stability testing.
- ✓ Fairness analysis.
- ✓ Data-quality testing.
- ✓ Target-leakage checks.
- ✓ Testing with unexpected or new categories.
- ✓ Comparison against HealthConnect business requirements.
- ✓ Documentation of remaining risks.